

Week 7

File Systems

IN 1011 Operating Systems

File Systems

- File Concept
- File Operations
- Disk and Directory Structure
- File-System Mounting
- File Sharing
- Protection

File Systems

- File Concept
- File Operations
- Disk and Directory Structure
- File-System Mounting
- File Sharing
- Protection

These are not file systems!

- File System refers to the logical structure (i.e. software) and not to the hardware in which the data is stored.



File Systems

- A file system allows data to be formatted, stored (e.g. on a hard disk) and accessed in a disciplined and convenient manner
- The users of a file system do not need to understand how the underlying physical storage media work
- The same file system can be implemented on many different devices (even in RAM)

File Attributes 1

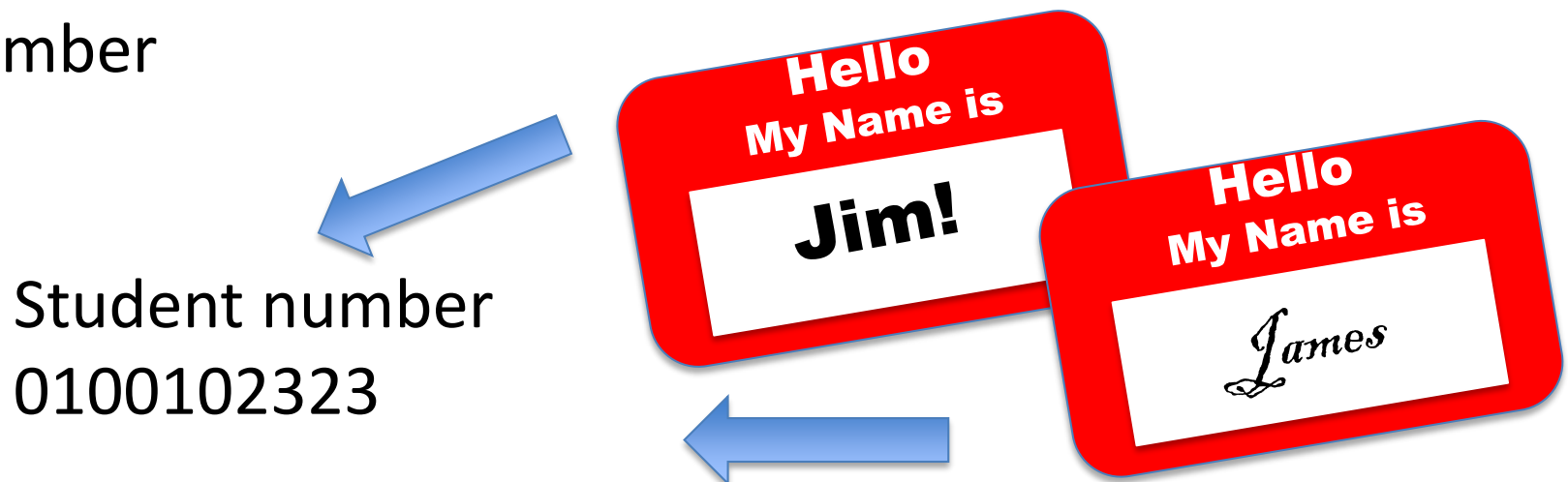
- Files contain data but also have other attributes:
 - **Name** – only information kept in human-readable form, this is what users see
 - **Identifier** – unique tag (number) which identifies each file within a file system (for Unix these are called i-numbers)

Student number
0100102323



File Attributes 2

- In Unix:
 - file names are only stored in directories, they are not attributes of files themselves
 - a directory contains a mapping from names to i-numbers
 - more than one name can be mapped to the same i-number



File Attributes 3

- Files contain data but also have other attributes:
 - **Type** – (in systems that support different file types)
 - **Location** – pointer to file location on device (e.g. disk)
 - **Size** – current file size
 - **Protection** – controls who can read, write, execute
 - **Time, date, and user identification** – data for ownership, protection, security, and auditing

Examples of File Info

Windows, Unix, Mac OS X



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EST 1894

```
Command Prompt
C:\Users\...\Desktop>dir
Volume in drive C is Windows
Volume Serial Number is 9805-82C8

Directory of C:\Users\...\Desktop

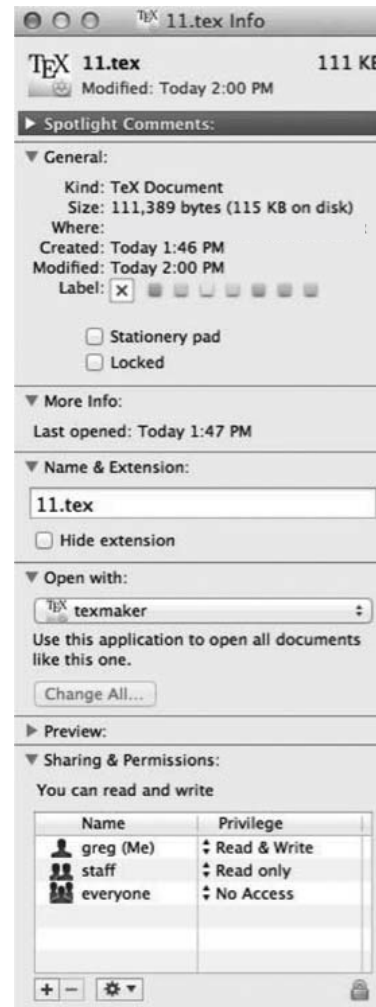
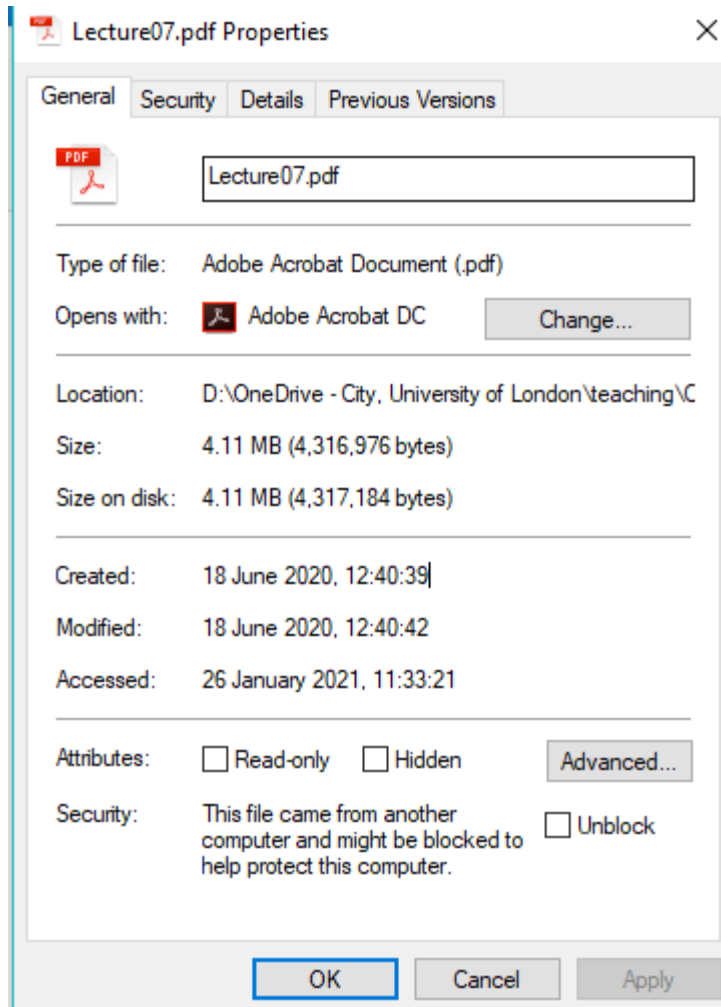
25/01/2021  16:22    <DIR>          .
25/01/2021  16:22    <DIR>          ..
27/07/2018  17:03                908 Anaconda Navigator.lnk
01/12/2020  10:37       748,968 Computer_2020_12_02_1037.docx
31/03/2020  08:59       328,067 CoverLetter_Macros.docx
09/10/2020  10:17       140,271 CS - Quickstart.pdf
10/05/2019  14:59       104,960 Expenses-Form-non-EOD-users.doc
22/09/2020  09:59       159,564 Fig1.png
27/10/2020  21:22         1,514 firefox.lnk
08/10/2020  08:46       165,169 firm2.jpg
13/08/2020  15:46         1,171 FreeMind.lnk
26/11/2020  12:45         2,146 GitHub Desktop.lnk
07/07/2018  17:19         1,227 HFSExplorer.lnk
10/11/2018  18:53         1,858 ImageMagick Display.lnk
10/08/2018  11:05         1,928 IrfanView 64 Thumbnails.lnk
10/08/2018  11:05         1,054 IrfanView 64.lnk
05/12/2018  16:46       13,792 Meeting_2018_12_05_City,KCL.docx
20/01/2021  09:46         2,377 Microsoft Teams.lnk
25/06/2020  14:54         595 OpenVPN Disconnect.lnk
15/05/2018  13:30       497,955 PalateCT.jpg
09/12/2020  09:41       154,896 PlanWSX343131.pdf
23/09/2020  08:09         21,165 PO 4500166967.pdf
18/10/2017  13:18         25,773 readNeutrophils2.m
20/05/2020  19:33         9,519 ...pdf
03/08/2020  10:11       58,458 Security_Exception_Request_Template_v5_0.docx
29/04/2020  09:49       63,412 SettingForExercises.jpg
10/04/2019  11:36         3,168 T00002.mat
09/10/2020  10:31       132,157 USDASC - MSci Data Science.pdf
27/10/2020  17:53       24,024 VK7 4501104469.pdf
01/12/2020  12:28         1,111 WinDirStat.lnk
07/06/2018  11:47         585 WriteAVideoToANAVIFileExample.m
05/10/2020  15:55         1,944 Zoom.lnk
          30 File(s)          2,669,736 bytes
           2 Dir(s)      9,567,694,848 bytes free

C:\Users\...\Desktop>

OpenSSH SSH client
^--bash-4.2$ ls
4D0E937EF974  css  EE1409  EE2701  EE3438  EEM300  IN1011  mail  Matlab  public
--bash-4.2$ ls -la
total 76876
drwx-----x  14          4096 Jan 22 16:07 .
drwxr-xr-x  649       16384 Jan 21 13:12 ..
-rwxrw----  1          4096 Jan 29 2016 ._2_XRay_CT_2.pptx
-rwxrw----  1          40 Jan 23 2017 4D0E937EF974
-rw-----  1          693 Jan 22 16:13 .bash_history
drwxrwxr-x  3          17 Dec 15 15:16 .cache
drwxrwxr-x  4          28 Dec 18 17:45 .config
-rw-----  1          489 Jul  1 2013 .cshrc
drwxr-xr-x  2          4096 Jul 26 2013 css
-rwxrw----  1          4096 Oct 17 2016 ._DS_Store
-rwxrw----  1          8196 Feb 17 2020 .DS_Store
drwxrwxr-x  3          4096 Jul 11 2019 EE1409
drwxrwxr-x  3          19 Jul 11 2019 EE2701
drwxrwxr-x  2          4096 Jul 11 2019 EE3438
drwxrwxr-x  3          4096 Feb  6 2020 EEM300
-rwxrw----  1          4096 Apr 11 2018 ._ElectroCardioGram.m
-rwxrw----  1          4096 Apr 11 2018 ._ElectroCardioGram.mlx
drwxrwxr-x  2          36 Jan 22 16:08 IN1011
-rw-----  1          452 Jul  1 2013 .login
-rw-----  1           6 Jul  1 2013 .logout
-rwxrw----  1          4096 Mar  8 2016 ._LookingInside_City.ppt
drwx-----  2          43 Jul  3 2013 mail
drwxrwxr-x  4          98 Jul 11 2019 Matlab
-rw-----  1          577 Aug  1 2013 .modulesbeginenv
-rw-----  1         11107 Jul  3 2013 .pine-debug1
-rw-----  1        14527 Jul  3 2013 .pinerc
drwx-----x 10          4096 Sep  3 13:01 public_html
-rwxrw----  1       39282605 Feb  9 2016 .smbdeleteAAA4c4a060
-rwxrw----  1       39282624 Feb  9 2016 .smbdeleteAAa6dd629
drwxrwxr-x  3          24 Oct 26 2015 .TemporaryItems
-rw-rw----  1          4096 Oct 26 2015 ._TemporaryItems
--bash-4.2$
```

Examples of File Info

Windows, Unix, Mac OS X



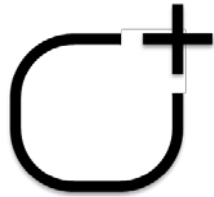
File Systems

- File Concept
- **File Operations**
- Disk and Directory Structure
- File-System Mounting
- File Sharing
- Protection

File Operations

- File is an abstract data type (i.e. a group of objects defined by its behaviour, the behaviour is defined by a values and operations)

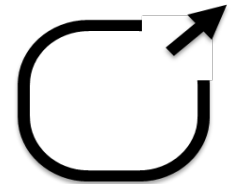
Create



Read



Open



Delete



Write



Close



File Operations 2

- Further operations
 - Reposition (a pointer) within file
 - Write – at write pointer location
 - Read – at read pointer location
 - Truncate
 - Append
 - Get attribute
 - Set attribute
 - Rename

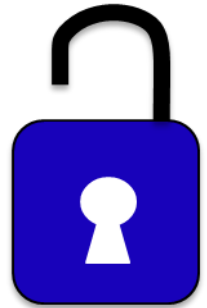


Open Files

- Several pieces of data are needed to manage open files:
 - **Open-file table:** tracks open files
 - **File pointer:** pointer to last read/write location, per *process* that has the file open
 - **File-open count:** counter of number of times a file is open – to allow removal of data from open-file table when last processes closes it
 - **Disk location of the file:** e.g. block on disk
 - **Access rights:** per-process access mode information

Open File Locking

- Provided by some operating systems and file systems
- Mediates access to a file
- Mandatory or advisory:
 - **Mandatory** – access is denied depending on locks held and requested
 - **Advisory** – processes can find status of locks and decide what to do

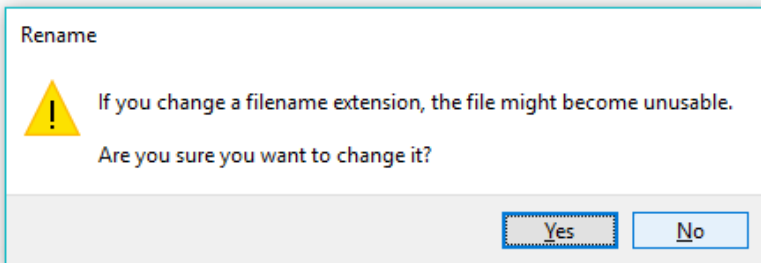


File Types

- An OS *may* distinguish between a range of different types
 - e.g. text, binary, sound, video
 - in such a typed system, copy binary to text would not be permitted (cf `int x = true`)
- Or (more common) all files are just a sequence of bytes

–OS and user applications may use *file name extensions* as **hints** about data formats

 Lecture_06_MemoryManagement.doc	✓	22/01/2021 09:49	Adobe Acrobat D...
 Lecture_06_MemoryManagement_Hand...	✓	22/01/2021 09:51	Adobe Acrobat D...



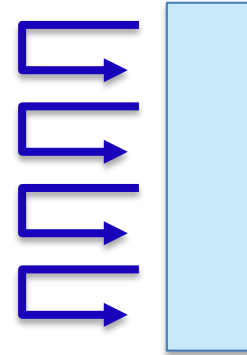
File Types – Name, Extension

file type	usual extension	function
executable	exe, com, bin or none	ready-to-run machine-language program
object	obj, o	compiled, machine language, not linked
source code	c, cc, java, pas, asm, a	source code in various languages
batch	bat, sh	commands to the command interpreter
text	txt, doc	textual data, documents
word processor	wp, tex, rtf, doc	various word-processor formats
library	lib, a, so, dll	libraries of routines for programmers
print or view	ps, pdf, jpg	ASCII or binary file in a format for printing or viewing
archive	arc, zip, tar	related files grouped into one file, sometimes compressed, for archiving or storage
multimedia	mpeg, mov, rm, mp3, avi	binary file containing audio or A/V information

Access Methods

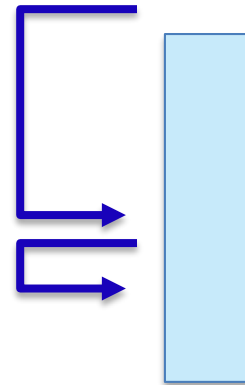
- Sequential Access

- read next
- write next
- reset

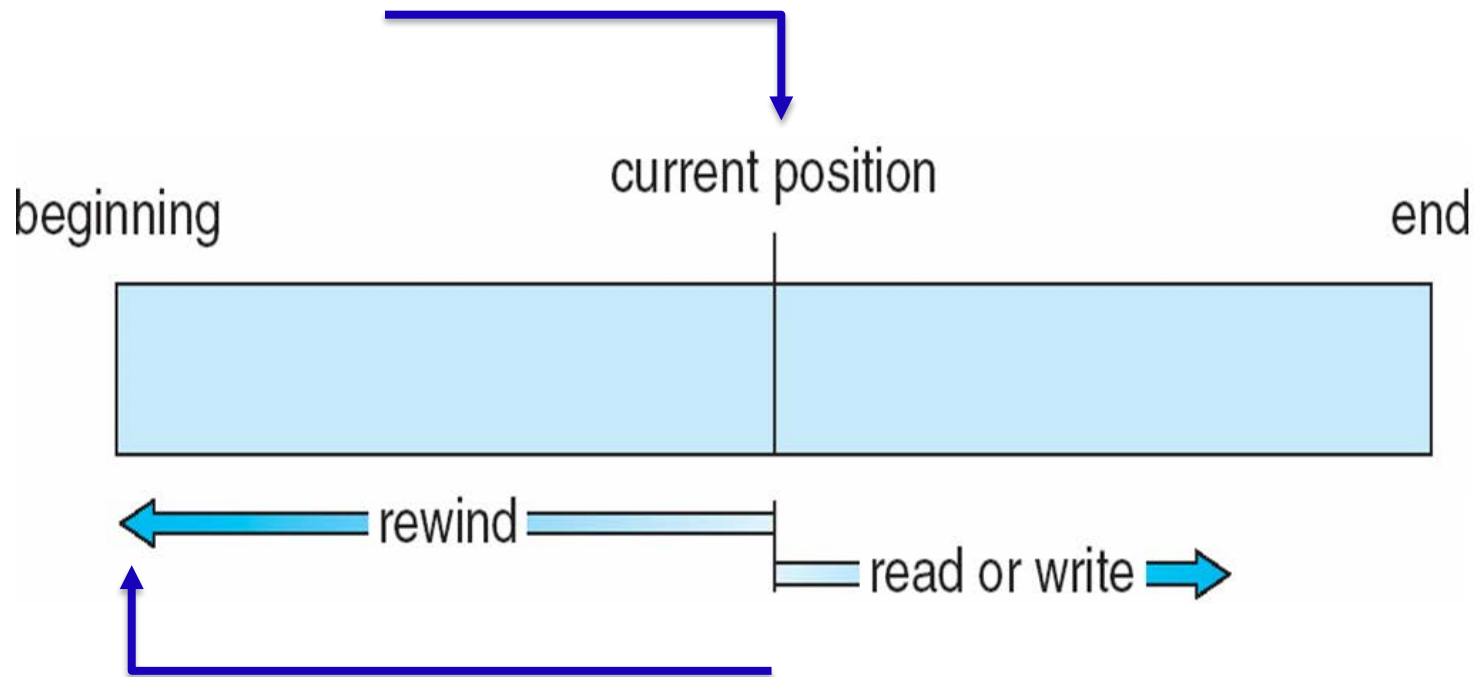


- Direct Access – file is fixed length logical records

- read *n*
- write *n*
- position to *n*
- read next
- write next



Sequential-access File



Simulation of Sequential Access on Direct-access File

sequential access	implementation for direct access
<i>reset</i>	<i>cp = 0;</i>
<i>read next</i>	<i>read cp;</i> <i>cp = cp + 1;</i>
<i>write next</i>	<i>write cp;</i> <i>cp = cp + 1;</i>

File Open and Read in C

```
#include <stdlib.h>
#define BUFSIZE 1024
#define O_RDONLY 00
int main ()
{
    char buffer [BUFSIZE];
    int length;
    int fd;
    if ( ( fd = open ("alice.txt", O_RDONLY) ) < 0 ) {
        perror("Unable to open data");
        exit (1);
    }
    while ( (length = read (fd, buffer, BUFSIZE)) > 0 ) {
        write (1, buffer, length);
    }
    close (fd);
    exit (0);
}
```

Standard library for I/O

Buffer Size

Predefined Read Only

Variables char, int

File Open and Read in C

```
#include <stdlib.h>
#define BUFSIZE 1024
#define O_RDONLY 00
int main ()
{
    char buffer [BUFSIZE];
    int length;
    int fd;
    if ( ( fd = open ("alice.txt", O_RDONLY) ) < 0 ) {
        perror("Unable to open data");
        exit (1);
    }
    while ( (length = read (fd, buffer, BUFSIZE)) > 0 ) {
        write (1, buffer, length);
    }
    close (fd);
    exit (0);
}
```

Syscall open returns a
file descriptor (positive int)

Name of the file

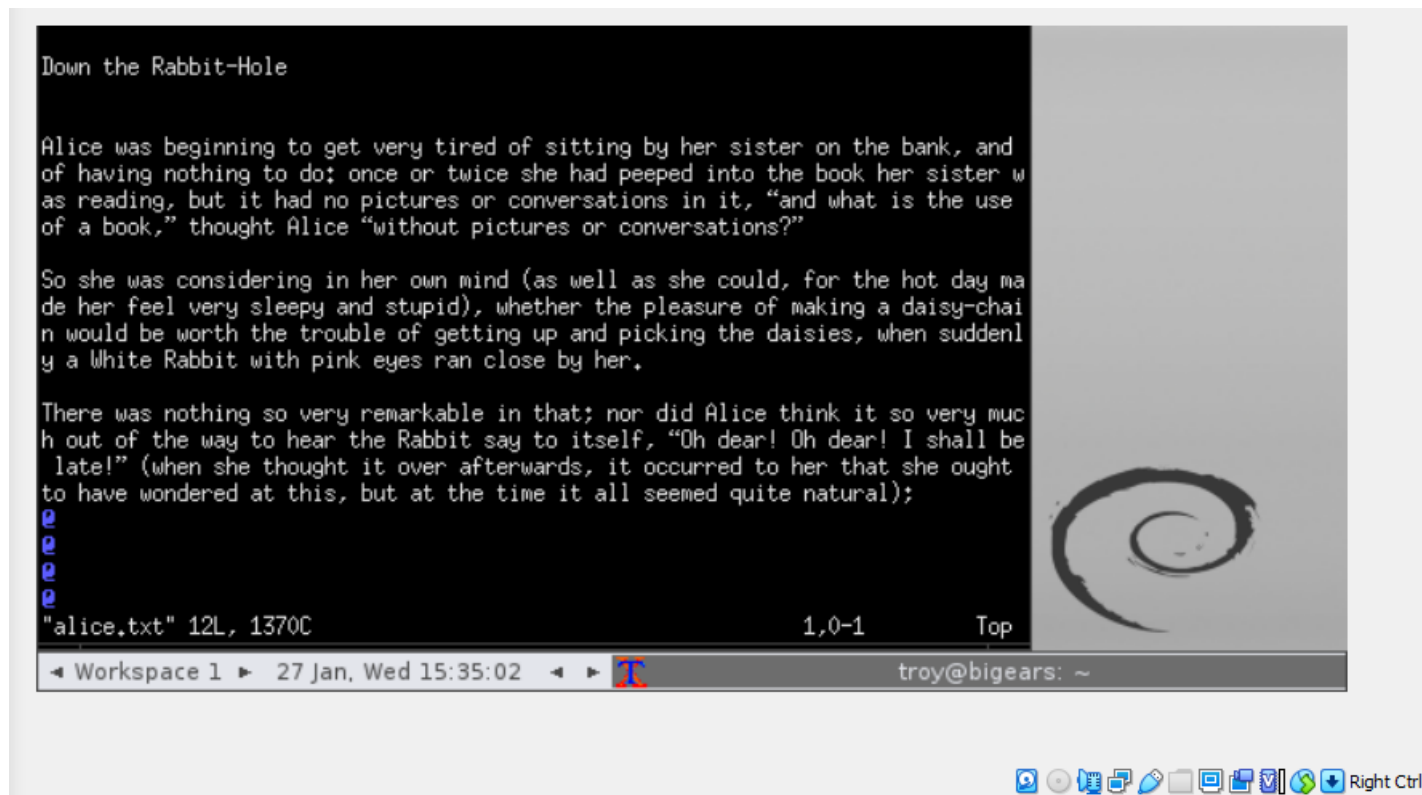
Predefined Read Only

Is this sequential or
direct access?

file descriptor 1 is the
“standard output”

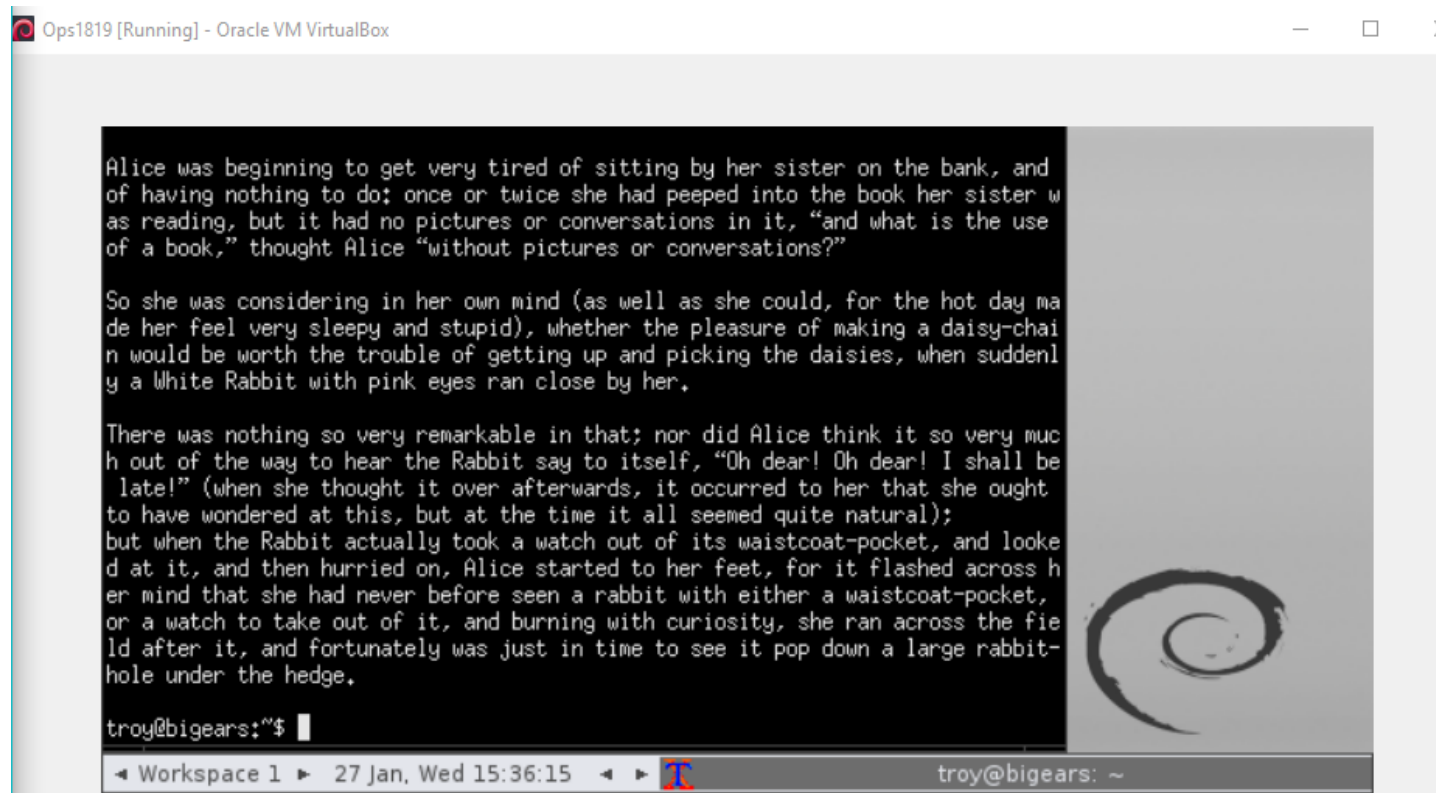
Extract from Alice in Wonderland

- <http://www.gutenberg.org/files/11/11-h/11-h.htm>



Run the file

```
gcc -o readFile readF.c  
./readFile
```



Ops1819 [Running] - Oracle VM VirtualBox

```
Alice was beginning to get very tired of sitting by her sister on the bank, and  
of having nothing to do; once or twice she had peeped into the book her sister w  
as reading, but it had no pictures or conversations in it, "and what is the use  
of a book," thought Alice "without pictures or conversations?"
```

So she was considering in her own mind (as well as she could, for the hot day ma
de her feel very sleepy and stupid), whether the pleasure of making a daisy-chai
n would be worth the trouble of getting up and picking the daisies, when suddenl
y a White Rabbit with pink eyes ran close by her.

There was nothing so very remarkable in that; nor did Alice think it so very muc
h out of the way to hear the Rabbit say to itself, "Oh dear! Oh dear! I shall be
late!" (when she thought it over afterwards, it occurred to her that she ought
to have wondered at this, but at the time it all seemed quite natural);
but when the Rabbit actually took a watch out of its waistcoat-pocket, and looke
d at it, and then hurried on, Alice started to her feet, for it flashed across h
er mind that she had never before seen a rabbit with either a waistcoat-pocket,
or a watch to take out of it, and burning with curiosity, she ran across the fie
ld after it, and fortunately was just in time to see it pop down a large rabbit-
hole under the hedge.

```
troy@bigears:~$
```

◀ Workspace 1 ▶ 27 Jan, Wed 15:36:15 troy@bigears: ~

Change the Buffer

Ops1819 [Running] - Oracle VM VirtualBox

```
#include <stdlib.h>
#include <string.h>
#define BUFSIZE 128
#define O_RDONLY      00
int main ()
{
    char buffer [BUFSIZE];
    int length;
    int fd;
    if ( ( fd = open ("alice.txt", O_RDONLY) ) < 0 ) {
        perror("Unable to open data");
        exit (1);
    }
    while ( (length = read (fd, buffer, BUFSIZE)) > 0 ) {
        write (1, buffer, length);
        write (1, " = ", 3);
    }
    close (fd);
    exit (0);
}
~
~
"readF2.c" 21L, 401C
```

1,0-1

All

◀ Workspace 1 ▶ 27 Jan, Wed 15:43:31



troy@bigears: ~

Change the Buffer

Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing to do: once or twice she had peeped into the book her sister was reading, but it had no pictures or conversations in it, "and what is the use of a book," thought Alice "without pictures or conversations?"

So she was considering in her own mind (as well as she could, for the hot day made her feel very sleepy and stupid), whether the pleasure of making a daisy-chain would be worth the trouble of getting up and picking the daisies, when suddenly a White Rabbit with pink eyes ran close by her.

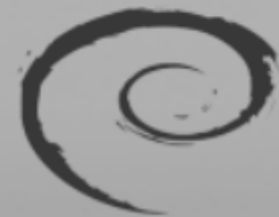
There was nothing so very remarkable in that; nor did Alice think it so very much out of the way to hear the Rabbit say to itself, "Oh dear! Oh dear! I shall be late!" (when she thought it over afterwards, it occurred to her that she ought to have wondered at this, but at the time it all seemed quite natural); but when the Rabbit actually took a watch out of its waistcoat-pocket, and looked at it, and then hurried on, Alice started to her feet, for it flashed across her mind that she had never before seen a rabbit with either a waistcoat-pocket, or a watch to take out of it, and burning with curiosity, she ran across the field after it, and was just in time to see it pop down a large rabbit-hole under the hedge.

troy@bigears:~\$

Workspace 1 27 Jan, Wed 15:37:28



troy@bigears: ~



File Open and Read in Java

```
static final int BUFSIZE = 1024;

public static void main(String[] args) {
    byte[] buffer = new byte[BUFSIZE];
    int length;
    InputStream is;
    try {
        is = new FileInputStream("data");
        while ( (length = is.read(buffer)) > 0 ) {
            System.out.write(buffer, 0, length);
        }
        is.close();
    } catch (IOException e) {
        System.err.println("Unable to open data: " + e);
        System.exit(1);
    }
}
```

File Systems

- File Concept
- File Operations
- Disk and Directory Structure
- File-System Mounting
- File Sharing
- Protection

Operations Performed on Directory

- Search for a file
- Create a file
- Delete a file
- List a directory
- Rename a file
- Traverse the file system



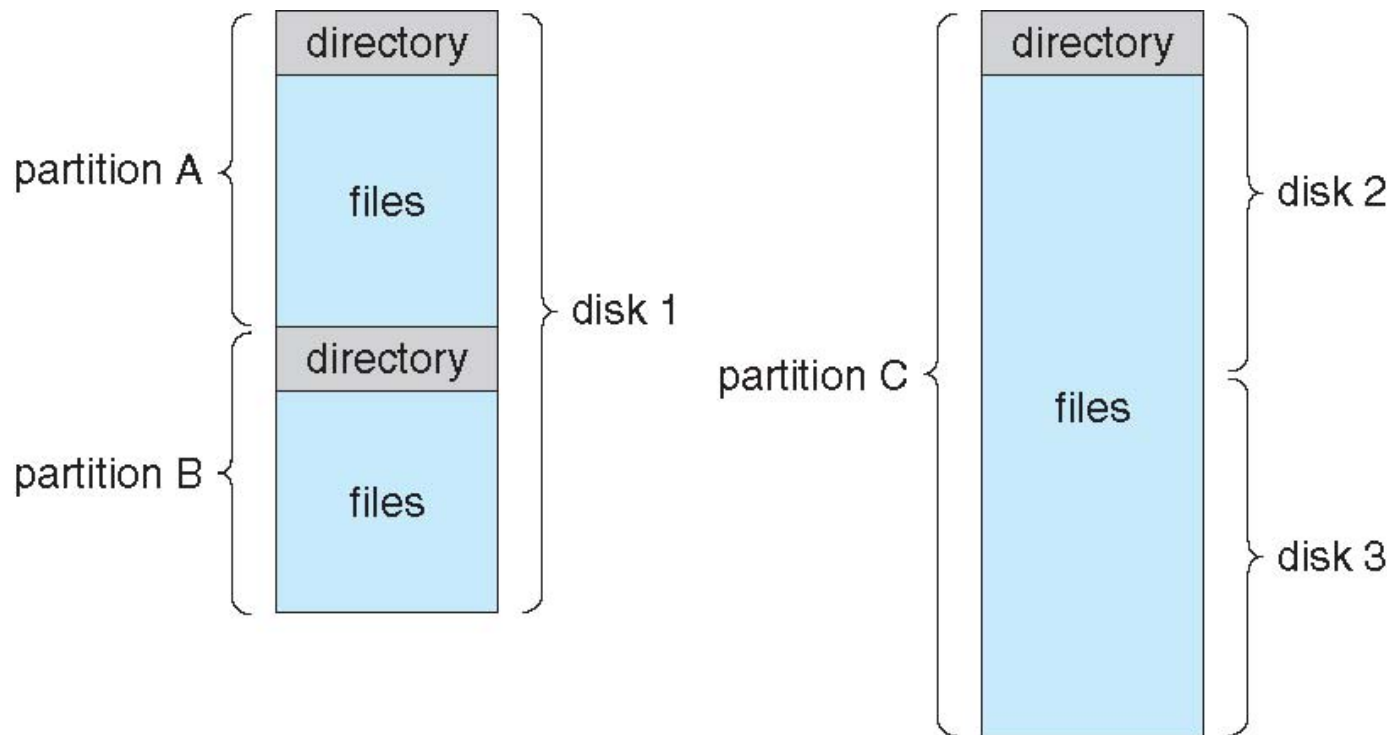
Disk Structure

- Disk can be subdivided into **partitions**
- Disk or partition can be used raw – without a file system, or **formatted** with a file system
 - Different partitions on the same disk may be formatted as different types of file system
 - Linux supports many: *ext2*, *ext3*, *ext4*, *ReiserFS*, ...

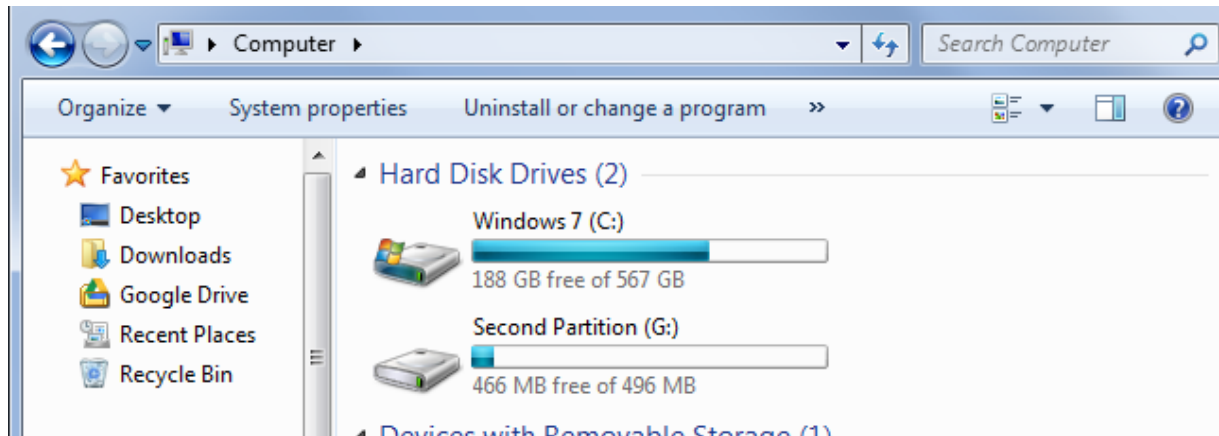


Disk Structure

- Each partition containing a file system will store info relating to that file system in a **device directory** or a **volume table of contents**



A Typical File-system Organization

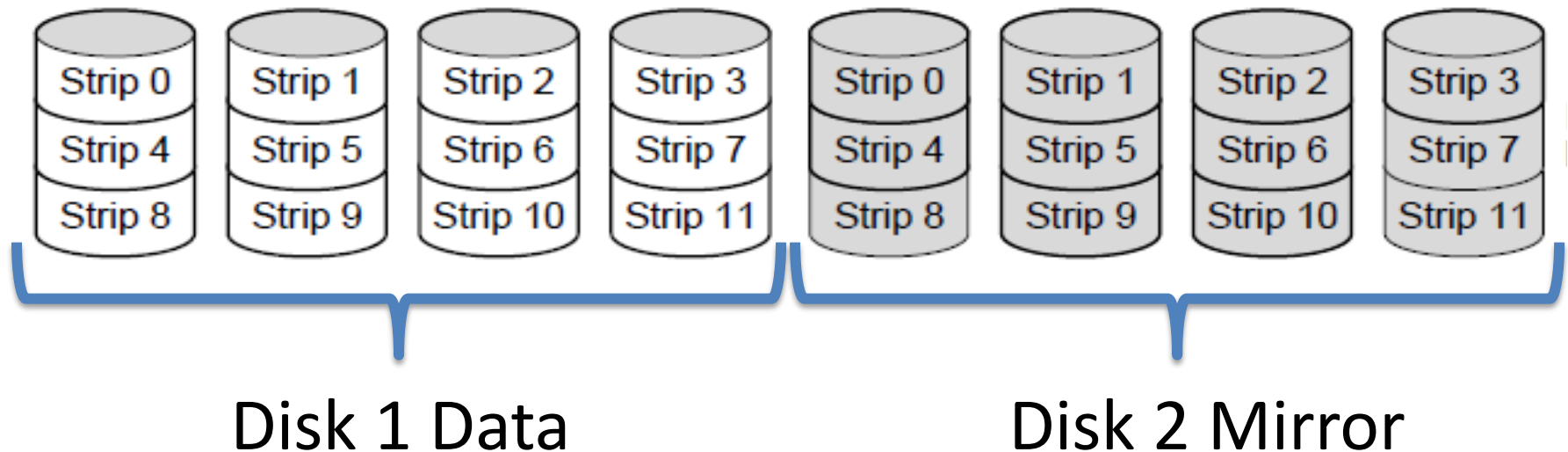


RAID

- Disks or partitions can be **RAID** protected against failure
- Redundant Array of Inexpensive (Independent) Disks
 - Use multiple smaller disks (c.f. one large disk)
 - Parallelism improves performance
 - Plus extra disk(s) for redundant data storage
- Provides fault tolerant storage system
 - Especially if failed disks can be “hot swapped”

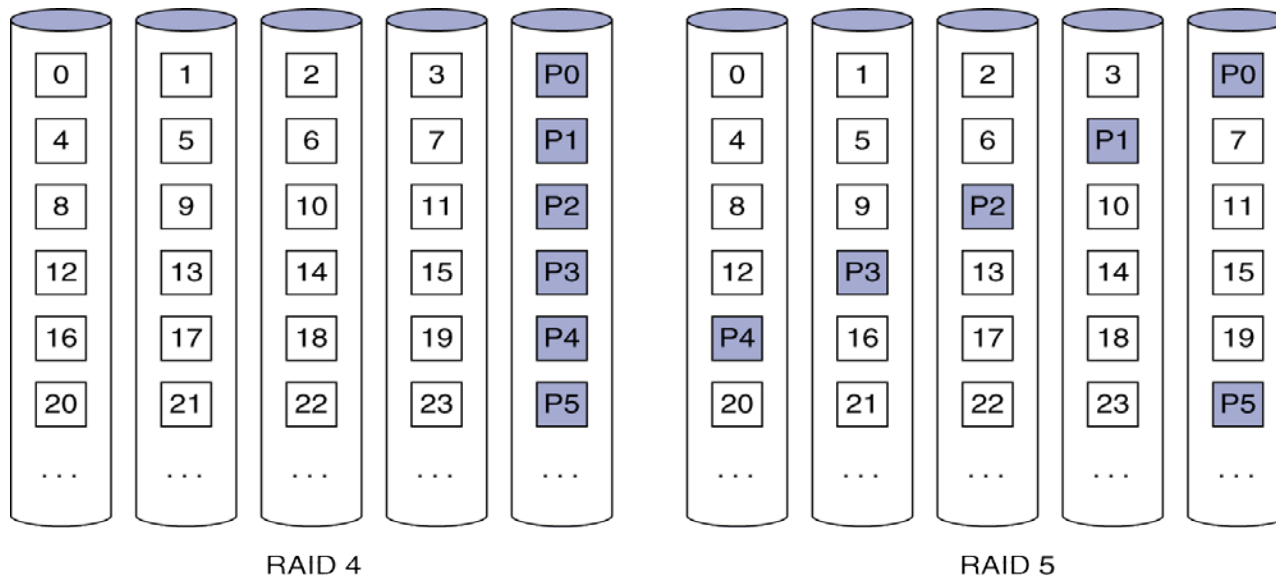
RAID 1

- RAID 1 Mirroring: **N + N disks**, replicate data
 - Write data to both data disk and mirror disk
 - On disk failure, read from mirror



RAID 5: Distributed Parity

- **N + 1 disks**
 - Data striped across N disks at block level
 - Redundant disk stores parity for a group of blocks, but parity blocks distributed across disks



Directory Organization

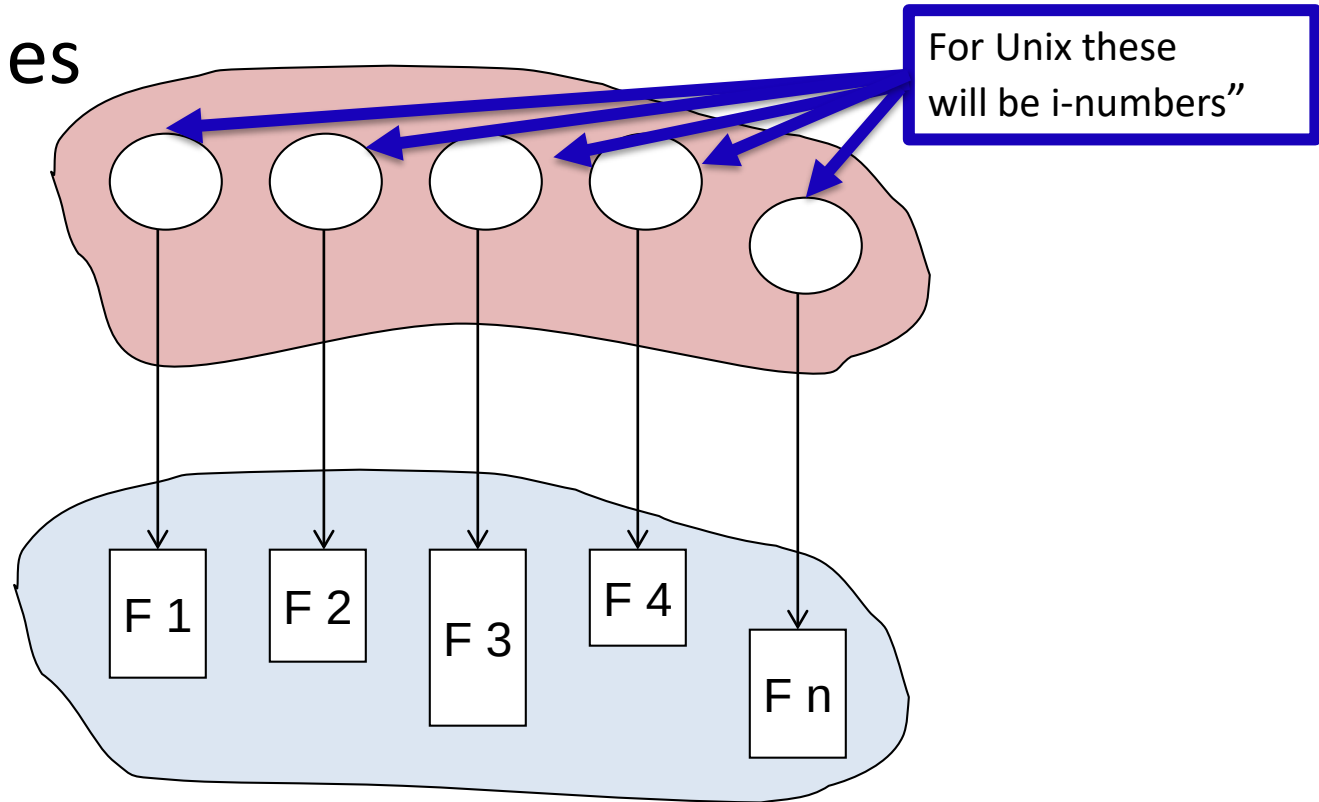
- The directory is organized logically to obtain
 - Efficiency – locating a file quickly
 - Naming – convenient to users
 - Two users can have same name for different files
 - The same file can have several different names
 - Grouping – logical grouping of files by properties, (e.g., all Java programs, all games, ...)

Directory Structure

- A collection of nodes containing information about all files

Directory
(Not a folder)

Files



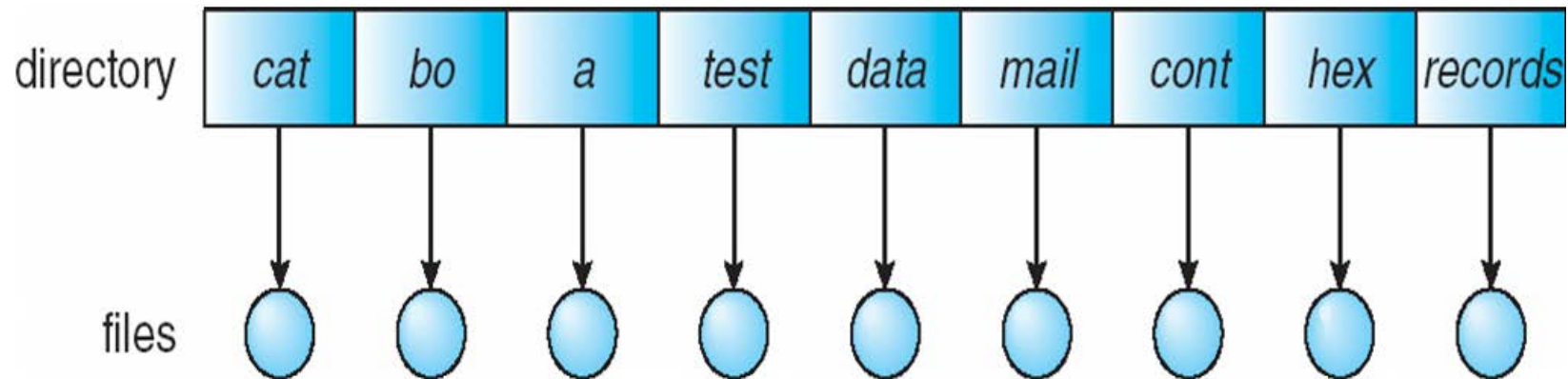
Both the directory structure and the files reside on disk

Additional File-system Structure

- Different file systems provide different ways of structuring file access to make life easier for users and applications
- From simple to sophisticated:
 - Single-level folder structure
 - Two-level folder structure
 - Tree-structured
 - Graph-structured

Single-Level Directory

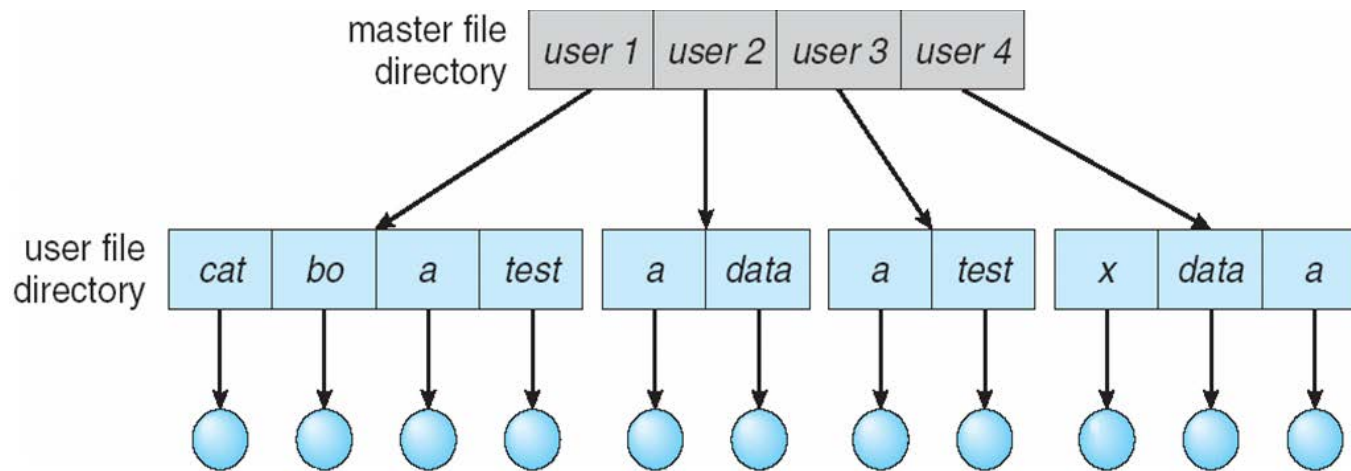
- A single directory for all users



- Naming problem
- Grouping problem

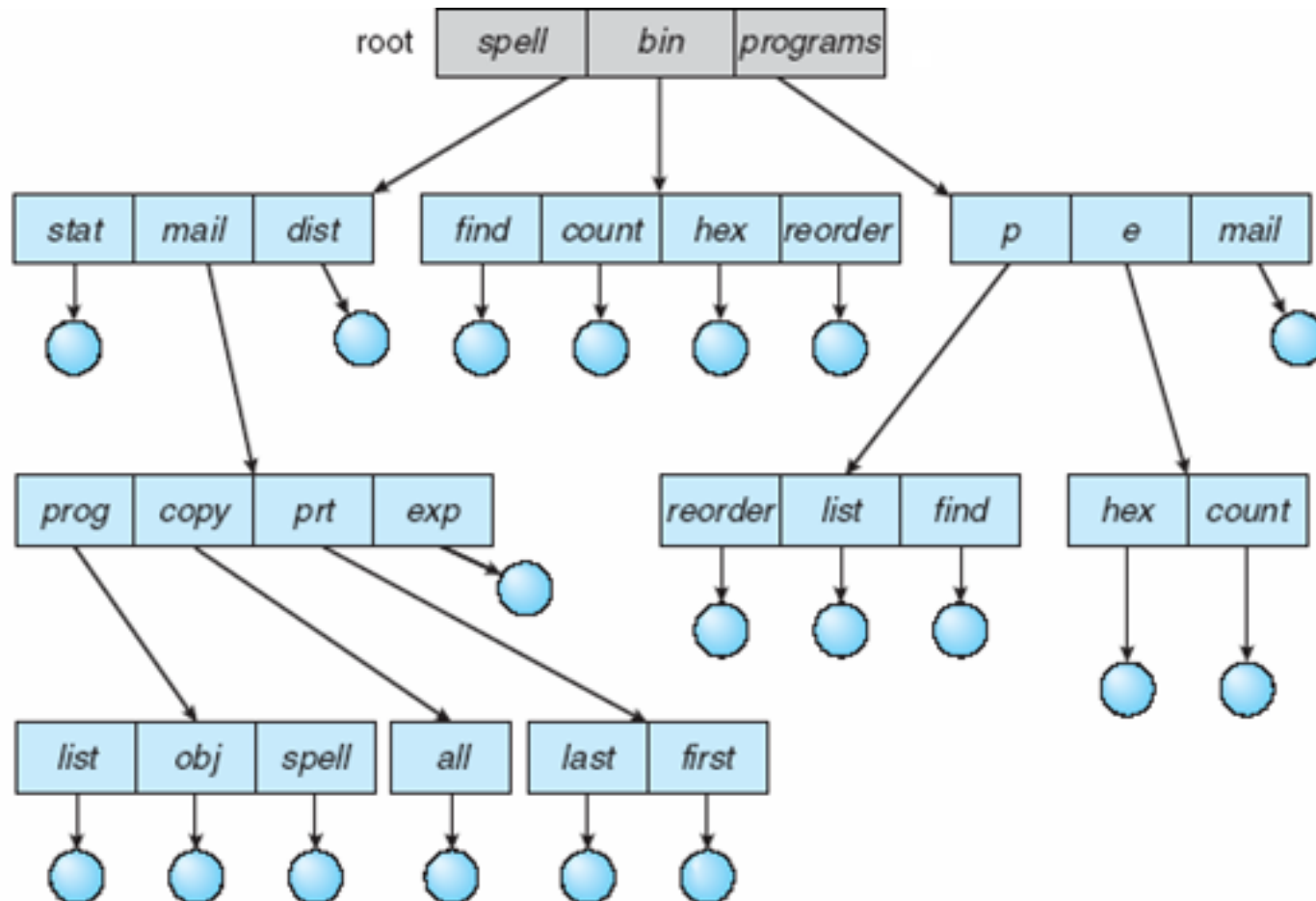
Two-Level Directory

- Separate directory for each user

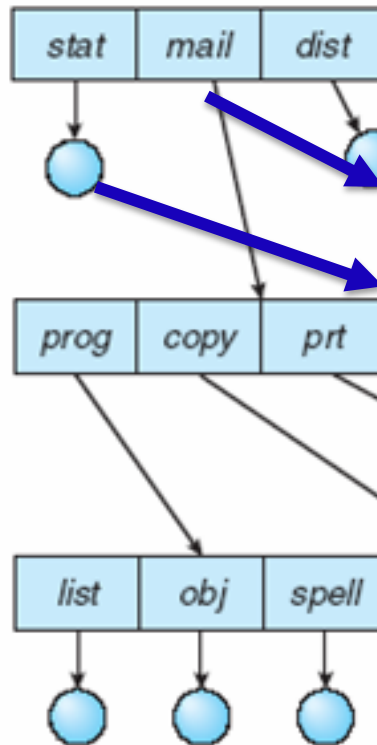


- Path name
- Can have the same file name for different user
- Efficient searching
- No grouping capability

Tree-Structured Directories



Tree-Structured Directories



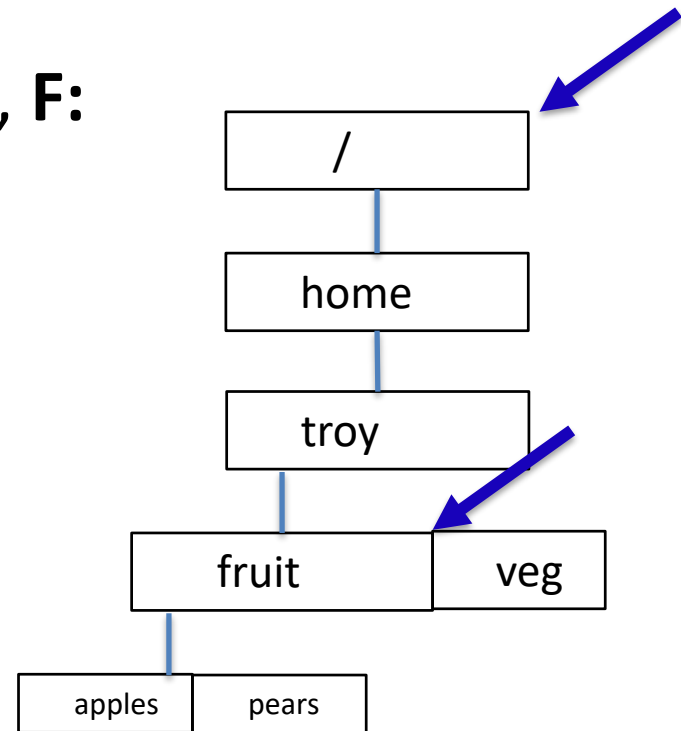
```
OpenSSH SSH client
-bash-4.2$ ls
4D0E937EF974 css EE1409 EE2701 EE3438 EEM300 IN1011 mail Matlab public_
-bash-4.2$ ls -la
total 76876
drwx-----x 14      103176    4096 Jan 22 16:07 .
drwxr-xr-x 649    root    root    16384 Jan 21 13:12 ..
-rwxrw---- 1      103176    4096 Jan 29 2016 ._2_XRay_CT 2.pptx
-rwxrw---- 1      103176    40 Jan 23 2017 4D0E937EF974
-rw----- 1      103176    693 Jan 22 16:13 .bash_history
drwxrwxr-x 3      103176    17 Dec 15 15:16 .cache
drwxrwxr-x 4      103176    28 Dec 18 17:45 .config
-rw----- 1      103176    489 Jul 1 2013 .cshrc
drwxr-xr-x 2      103176    4096 Jul 26 2013 css
-rwxrw---- 1      103176    4096 Oct 17 2016 ._DS_Store
-rwxrw---- 1      103176    8196 Feb 17 2020 .DS_Store
drwxrwx--- 3      103176    4096 Jul 11 2019 EE1409
drwxrwx--- 3      103176    19 Jul 11 2019 EE2701
drwxrwx--- 2      103176    4096 Jul 11 2019 EE3438
drwxrwx--- 3      103176    4096 Feb 6 2020 EEM300
-rwxrw---- 1      103176    4096 Apr 11 2018 ._ElectroCardioGram.m
-rwxrw---- 1      103176    4096 Apr 11 2018 ._ElectroCardioGram.mlx
drwxrwxr-x 2      103176    36 Jan 22 16:08 IN1011
-rw----- 1      103176    452 Jul 1 2013 .login
-rw----- 1      103176    6 Jul 1 2013 .logout
-rwxrw---- 1      103176    4096 Mar 8 2016 ._LookingInside_City.ppt
drwx----- 2      103176    43 Jul 3 2013 mail
drwxrwx--- 4      103176    98 Jul 11 2019 Matlab
-rw----- 1      103176    577 Aug 1 2013 .modulesbeginenv
```

Tree-Structured Directories (Cont.)

- Efficient searching
- Grouping Capability
- Current directory (working directory)
 - `$ nedit /home/troy/WebServer.java`
- OR
 - `$ cd /home/troy`
 - `$ nedit WebServer.java`

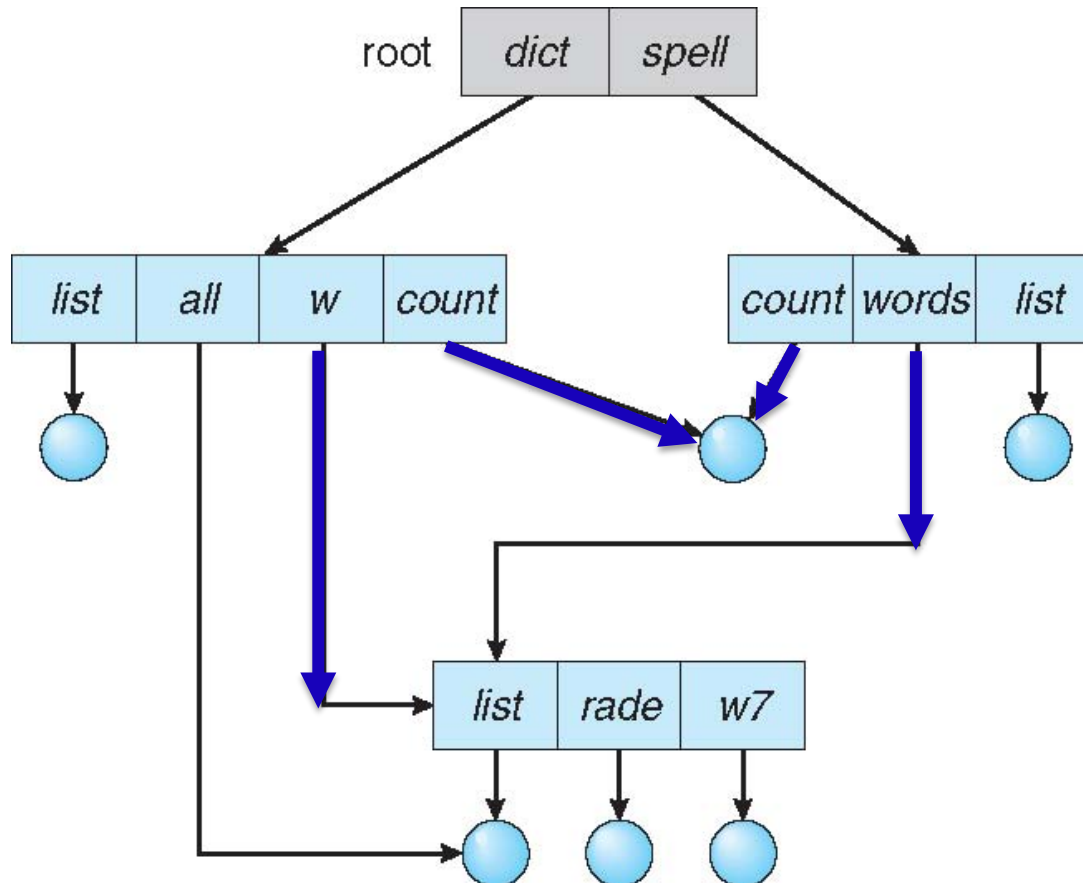
Tree-Structured Directories (Cont)

- **Absolute** or **relative** path name
 - Unix absolute path starts with **/**
 - Windows absolute path starts **C:**, **F:**
- **Example**
 - `mkdir /home/troy/fruit`
 - `cd /home/troy/fruit`
 - `touch apples`
 - `touch pears`
 - `mkdir ../veg`



Acyclic-Graph Directories

- Have shared subdirectories and files



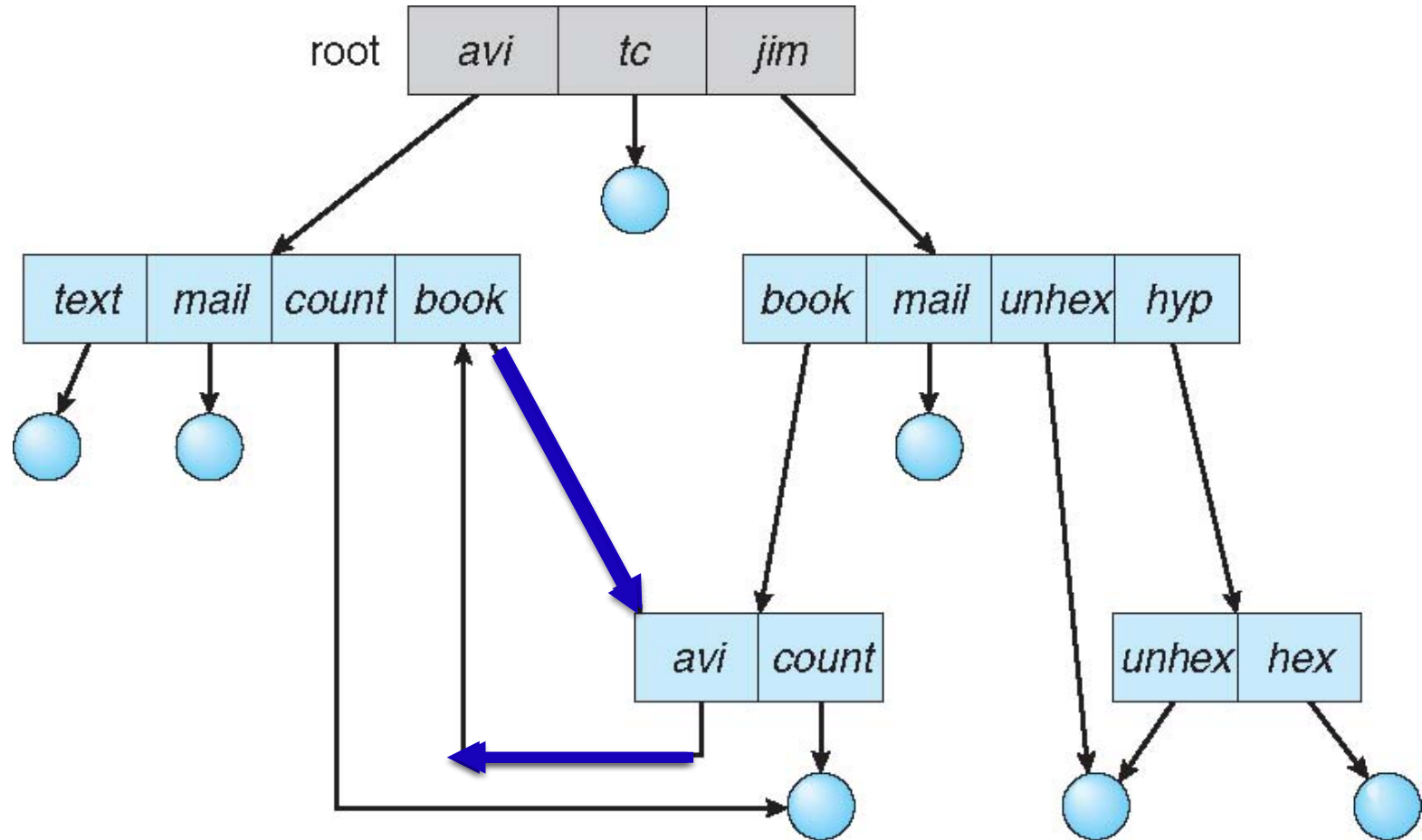
Acyclic-Graph Directories

- Two different names (aliasing)
- If **dict** deletes **count** $\Rightarrow$ dangling pointer
- Various more or less complicated solutions:
 - Unix: only delete the file (inode) when last ref (name) is removed
 - File system must keep track of number of references

Acyclic-Graph Directories

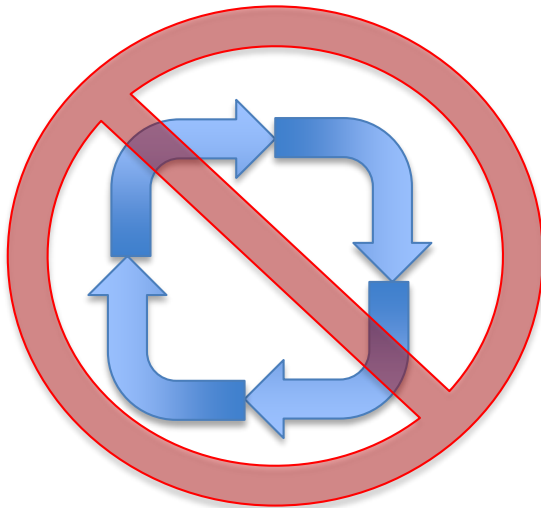
- Two different names (aliasing)
- If **dict** deletes **count** $\Rightarrow$ dangling pointer
- New directory entry type: ***symbolic link***
 - really just another file containing a string which is a path (pointer) to an existing file
 - Resolve the link – follow pointer to locate the file
 - Deleting the link has no effect on the file or vice versa

General Graph Directory



General Graph Directory (Cont.)

- How do we guarantee no cycles?
 - Allow only links to file not subdirectories
 - Every time a new link is added use a cycle detection algorithm to determine whether it is OK



File Systems

- File Concept
- File Operations
- Disk and Directory Structure
- **File-System Mounting**
- File Sharing
- Protection

File System Mounting

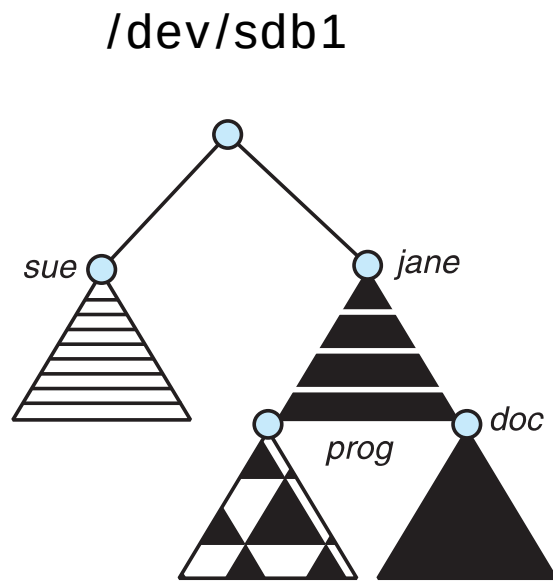
- A file system must be **mounted** before it can be accessed
- Mounting makes the file system available at a **mount point** within a previously mounted file system
- Need some special mechanism to create the root file system

File System Mounting

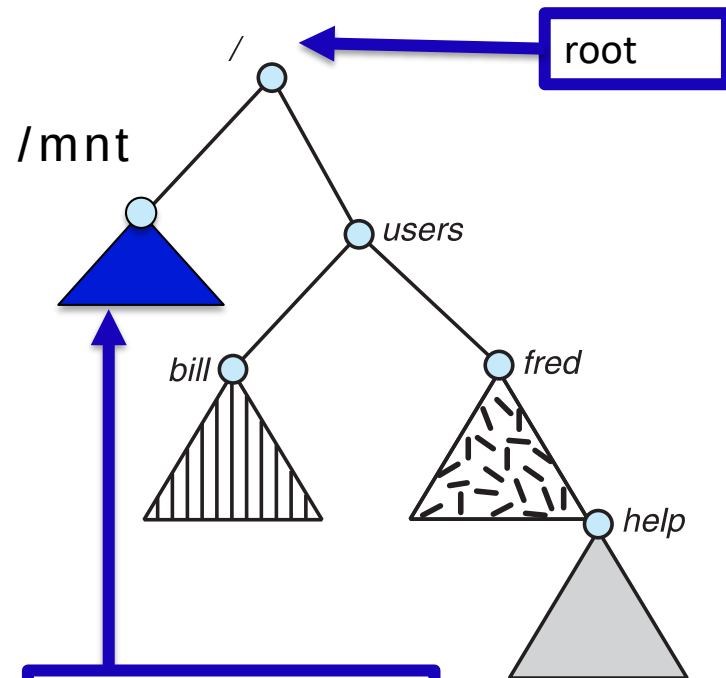
```
$ ssh @linux.city.ac.uk
linux.city.ac.uk's password:
Last login: 2015 from
-bash-4.1$ mount
/dev/mapper/system-root on / type ext4 (rw)
proc on /proc type proc (rw)
sysfs on /sys type sysfs (rw)
devpts on /dev/pts type devpts (rw,gid=5,mode=620)
tmpfs on /dev/shm type tmpfs (rw)
/dev/sda1 on /boot type ext4 (rw)
/dev/mapper/system-home on /home type ext4 (rw)
/dev/mapper/system-var on /var type ext4 (rw)
none on /proc/sys/fs/binfmt_misc type binfmt_misc (rw)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw)
swansea:/u1.swansea on /u1.swansea type nfs (rw,soft,sloppy,addr=10.200.34.198)
swansea:/u4.swansea on /u4.swansea type nfs (rw,soft,sloppy,addr=10.200.34.198)
truro:/net/linux on /net/linux type nfs (ro,soft,sloppy,addr=10.200.34.7)
-bash-4.1$ pwd
/u4.swansea/s15
-bash-4.1$
```

File System Mounting

- Illustration of mounted and unmounted file systems



An unmounted system



A mounted system

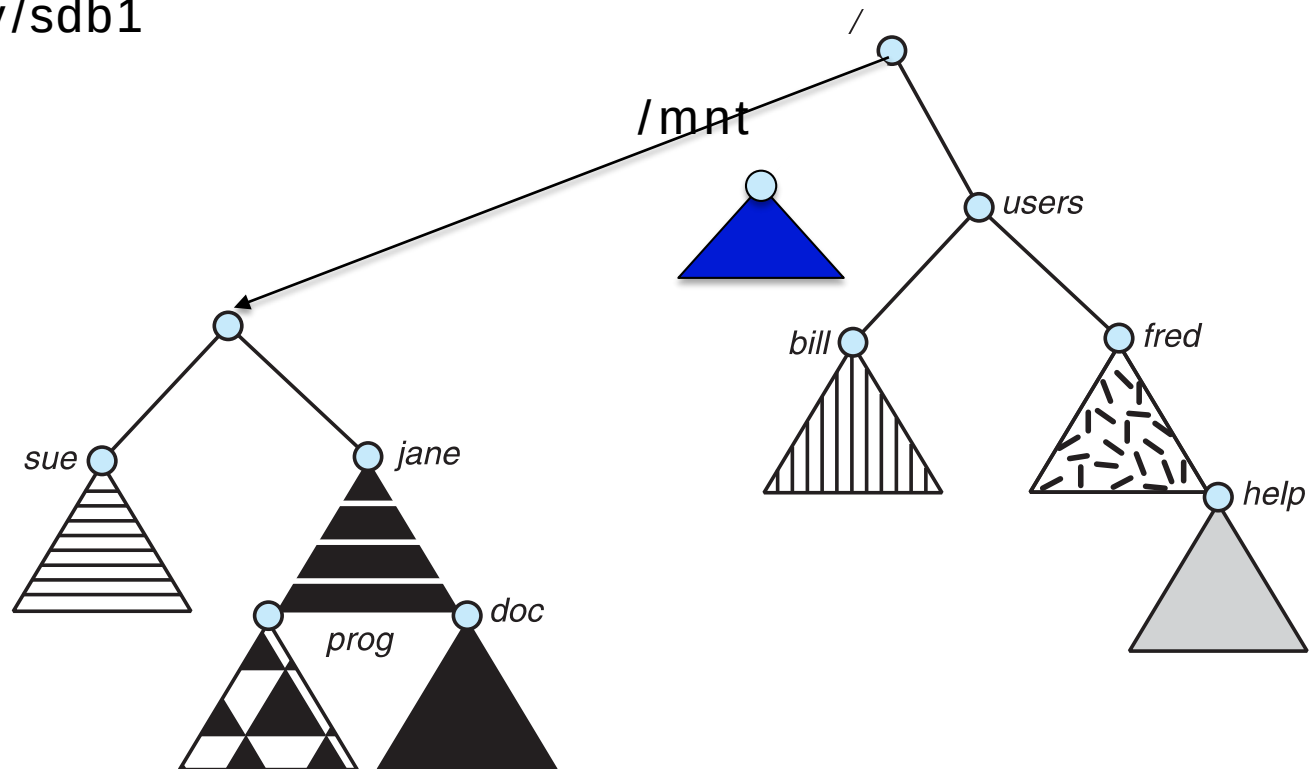
File System Mounting

```
troy@bigears: ~  
troy@bigears:~$  
troy@bigears:~$ su  
Password:  
root@bigears:/home/troy# mount -t ext4 /dev/sdb1 /mnt  
root@bigears:/home/troy#
```

File System Mounting

- Illustration of mounted and unmounted file systems

/dev/sdb1



File Systems

- File Concept
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- Protection

File Sharing

- Sharing of files on multi-user systems is desirable
- Sharing may be done through a **protection scheme**
- On distributed systems, files may be shared across a network
- Network File System (NFS) is a common distributed file-sharing method



File Sharing

- If multi-user system
 - User IDs identify users, allowing permissions and protections to be per-user
 - Group IDs allow users to be in groups, permitting group access rights
 - **Owner** of a file / directory
 - **Group** of a file / directory

File Sharing – Remote File Systems

- Uses networking to allow file system access between systems
 - Manually via programs like FTP
 - Automatically, seamlessly using distributed file systems
 - Semi automatically via the world wide web



File Systems

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- Protection

Protection

- Operating system consists of a collection of 'objects': hardware or software
- Each object has a **unique name** and can be accessed through a well-defined **set of operations**
- The Protection problem:
 - *Ensure that each object is accessed correctly and only by those processes that are allowed to do so.*



Protection

- File owner/creator should be able to control:
 - what can be done
 - by whom
- Types of access
 - Read
 - Write
 - Execute
 - List and access children in the directories
 - Delete

Access Lists and Groups

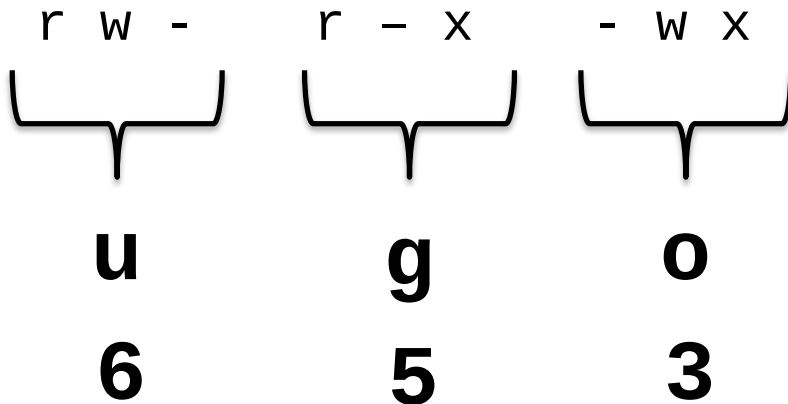
- Three mode of access: read (**r**), write (**w**), execute (**x**)

0	-	-	-	means no permissions for anything	0	0	0
1	-	-	x	means only execute	0	0	1
2	-	w	-	means only write	0	1	0
3	-	w	x	means write and execute	0	1	1
4	r	-	-	means only read	1	0	0
5	r	-	x	means read and execute	1	0	1
6	r	w	-	means read and write	1	1	0
7	r	w	x	means read, write and execute	1	1	1

- Notice the order in which these are listed

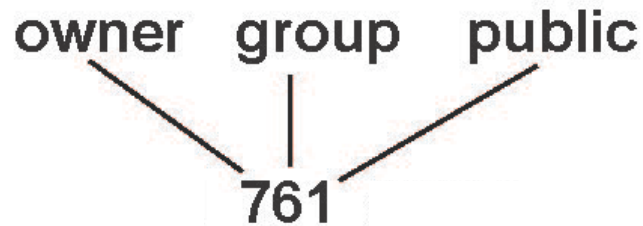
Access Lists and Groups

- Three classes of user: owner (**u**), group (**g**), other (**o**)
- There are permission for each of these users in the same order



Access Lists and Groups

- Example



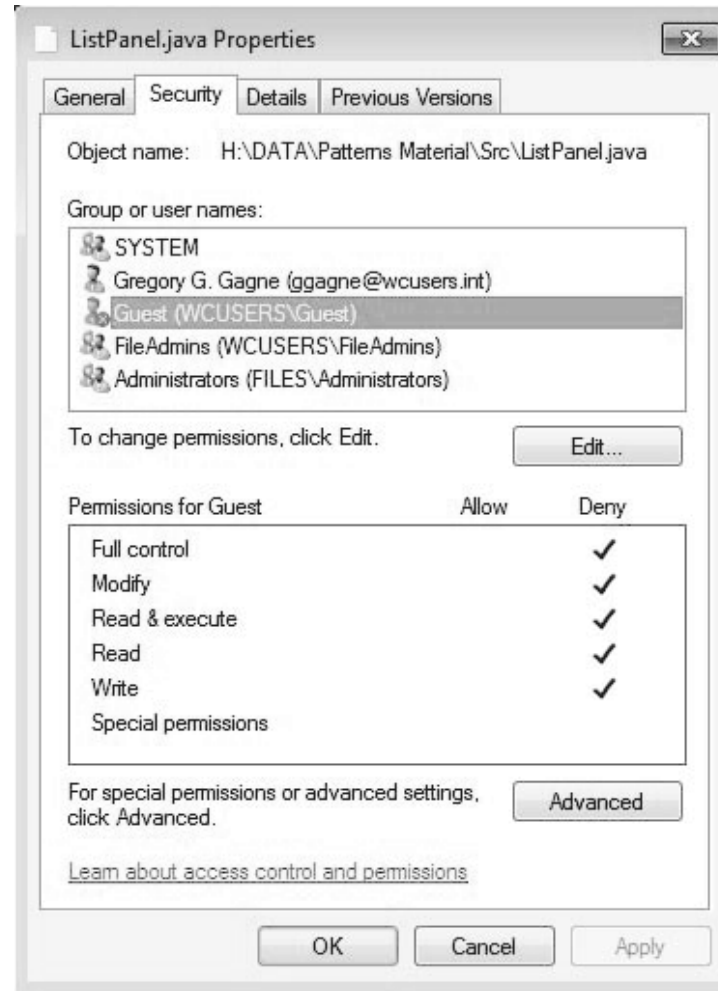
- a) owner access 7 $\Rightarrow$ 1 1 1
- b) group access 6 $\Rightarrow$ 1 1 0
- c) public access 1 $\Rightarrow$ 0 0 1

- Or

– 111 110 001

```
-bash-4.2$ ls -la
total 76876
drwx-----x 14 sb
drwxr-xr-x 649 ro
-rwxrw---- 1 sb
-rwxrw---- 1 sb
-rw----- 1 sb
drwxrwxr-x 3 sb
drwxrwxr-x 4 sb
-rw----- 1 sb
drwxr-xr-x 2 sb
-rwxrw---- 1 sb
-rwxrw---- 1 sb
drwxrwx--- 3 sb
drwxrwx--- 3 sb
drwxrwx--- 2 sb
drwxrwx--- 3 sb
-rwxrw---- 1 sb
-rwxrw---- 1 sb
drwxrwxr-x 2 sb
```

Windows / Mac Access-Control List Management



A Sample UNIX Directory Listing

file	user	group				
-rw-rw-r--	1 pbg	staff	31200	Sep 3 08:30	intro.ps	
drwx-----	5 pbg	staff	512	Jul 8 09:33	private/	
drwxrwxr-x	2 pbg	staff	512	Jul 8 09:35	doc/	
drwxrwx---	2 pbg	student	512	Aug 3 14:13	student-proj/	
-rw-r--r--	1 pbg	staff	9423	Feb 24 2003	program.c	
-rwxr-xr-x	1 pbg	staff	20471	Feb 24 2003	program	
drwx--x--x	4 pbg	faculty	512	Jul 31 10:31	lib/	
drwx-----	3 pbg	staff	1024	Aug 29 06:52	mail/	
drwxrwxrwx	3 pbg	staff	512	Jul 8 09:35	test/	

folder

A Sample UNIX Directory Listing

In this example the group names and user names are the same, but a group is *not* a user (Linux has created a default group for each user).

folder

User/
owner

group

```
drwxr-xr-- 2 troy troy 4096 Mar 25 21:31 dir
-r--r--r-x 1 troy troy 2448 Mar 25 21:32 file1
--w--w-r-- 1 zapata zapata 3273 Mar 25 21:33 file2
-rw-r--r-- 1 root root 11266 Mar 25 21:33 file3
-rwxr-xr-- 1 troy troy 22 Mar 25 22:01 file4
```

file

A Sample UNIX Directory Listing

```
drwxr-xr-- 2 troy troy 4096 Mar 25 21:31 dir
-r--r--r-x 1 troy troy 2448 Mar 25 21:32 file1
--w--w-r-- 1 zapata zapata 3273 Mar 25 21:33 file2
-rw-r--r-- 1 root root 11266 Mar 25 21:33 file3
-rwxr-xr-- 1 troy troy 22 Mar 25 22:01 file4
```

	file1	file2	file3	file4	dir
troy	read	read	read	read write execute	read write execute
zapata	read execute	write	read	read	read
helen	read execute	read	read	read	read

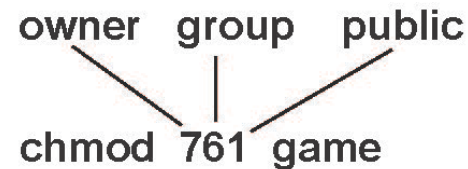
Unix: directories and the x bit

- Think of a directory as a special file containing a list of file names
 - You need **r** access to *list* the contents
 - You need **w** access to *change* the contents (add a file, rename a file or delete a file from the directory)
- The “execute” bit for directories is special:
 - You need **x** access to “traverse” a directory in order to access the files within it
 - If you have **x** but not **r** you can still use the files within *if you know their names* (`ls` can't help you)

Access Lists and Groups

- Users can change permissions for a particular file (say game) or subdirectory, define an appropriate access.

```
$chmod 761 game
```



- A manager or root user can to create and modify the owner and group of a file, also add users to the group.

- Change/attach group:

```
$root $ chgrp G game
```

- Change owner and group of a file

```
$root $ chown user1:group2 game
```

Access Matrix: Mechanism vs Policy

- **Access matrix** design separates mechanism from policy
 - Mechanism
 - OS provides matrix structure + rules
 - OS ensures that the matrix is only manipulated by authorized agents and strictly enforces rules
- **Policy**
 - Users dictate policy by setting entries in the matrix to specify who can access what

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